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SE_SPnet: Rice leaf disease prediction using stacked parallel convolutional neural network with squeeze-and-excitation

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Abstract

Rice is one of the significant crops, and the early identification and prevention of its diseases are essential to ensure adequate and healthy availability to the world's growing population. The use of image processing is an encouraging method for automatic rice leaf disease identification and detection. In particular, the recent advancements indicate the effectiveness of convolutional neural network (CNN) based deep learning approaches. In this direction, the present work proposes a novel stacked parallel convolution layers-based network (SPnet) with the squeeze-and-excitation (SE) architecture, named (SE_SPnet), for classifying diseased rice leaf images. The stacked parallel network block comprises four parallel convolution layers with different kernel sizes for abstractions of the global and local features. The SE block extracts feature information automatically while removing invalid ones. We compare the SE_SPnet model with state-of-the-art CNN models such as VGG16, DenseNet121, and InceptionV3 based on computational effort, accuracy, sensitivity, specificity, precision, recall, and F1-score. The experimental results show that the SE_SPnet outperforms standard CNN models for the considered rice leaf disease image datasets. In particular, the SE_SPnet achieves the highest accuracy (99.2%), sensitivity (98.2%), specificity (98.5%), precision (98.4%), recall (98.2%), and F1-score (98.5%) while using stochastic gradient descent (with momentum) optimizer with a 0.01 learning rate. Furthermore, the SE_SPnet also exhibits to outperform when compared with some of the most recent and relevant existing works.

KEYWORDS
CNN, deep learning, leaf disease, smart agriculture, squeeze-and-excitation

1 | INTRODUCTION

Agriculture and the related sectors have been at the core of the Indian economy. Over 70% of rural households depend on agriculture, and 54.6% of the workforce across the board is involved in agriculture and its associated sectoral activities (Census 2011). Moreover, the contribution of agriculture to the Indian GDP increased from 17.8% in 2019–20 to 19.9% in 2020–21. From a socio-economic perspective, agriculture is a sector that needs special attention and awareness across the globe. Therefore, there is a need for technological intervention in the agriculture sector to make it smart agriculture. This will help to improve agricultural productivity and raise farmers' income. Smart farming is found to be more effective than the conventional approach to farming. Smart farming is now assisting in crop disease detection as well. For example, the use of artificial intelligence (AI) and the internet of things (IoT) can help monitor the crop life cycle and detect various diseases. Using smart farming applications, farmers can easily monitor crops from anywhere. India's marginal and small-scale farmers struggle to combine physical and digital infrastructures, which severely restricts their ability to increase their revenue. As per the National Association of Software and Services Companies (NASSCOM) report published in 2019, more than 450 agri-based, technology-driven start-ups are located in India (Singh et al., 2020). Such agri-based tech-driven start-ups have a great scope to provide cost-effective solutions to small and marginal farmers and help them to improve their productivity and income through technology interventions.

In India, 107.04 million tonnes of Kharif rice are expected to be produced overall between 2021 and 2022. As reported by the Ministry of Agriculture & Farmers Welfare, Govt. of India (Ministry of Agriculture & Farmers Welfare, 2022), compared to the average production during the previous 5 years, which was 97.83 million tonnes, the average production has increased by 9.21 million tonnes. In India and around the world, rice is a common food. More than half of the world's population consumes rice as food (Qin & Zhang, 2005), and this number is likely to rise in the future (Johannes et al., 2017). In the past few years, the agriculture sector has faced numerous challenges. In particular, the rice crop suffers from diverse diseases, which harms the productivity and quality of the yield. The yield of rice could be decreased by 10%–30% as a result of certain diseases (Singh et al., 2019).

Rice leaf diseases are usually based on the appearance of the leaves. The rice leaf disease has colour and texture symptoms, and these diseases can be identified based on these attributes. These diseases are often visible to the farmers' naked eye by manually examining the leaves and can be diagnosed based on their own experience with their senses. This process is costly, slow, and non-scalable. As a result, technological intervention is a need of the hour to assist farmers in detecting these diseases at the initial stage.

Several revolutionary developments have been made to resolve this problem. With the rapid growth of technologies, various approaches like machine learning approach and computer vision have widely been proposed by investigators in crop disease detection (Liu et al., 2017; Pooja et al., 2017). Conventional image processing techniques provide solutions but have limitations in terms of scalability, performance, efficiency, and detection accuracy. Deep learning has emerged in the field of computer vision as a means of overcoming these limitations, particularly in the areas of image classification and object detection, which can assist farmers and researchers in accurately and efficiently identifying the different types of rice leaf diseases. An extremely effective deep-learning network is convolutional neural network (CNN). It employs an end-to-end structure and discards complicated image preprocessing and feature extraction tasks. This also significantly enhances the process of disease identification (Boulent et al., 2019; Chao et al., 2020; Ma et al., 2018). Presently, the automatic prediction of leaf diseases system integrated with CNN is beneficial to attain high correctness and precision.

Motivated by this, we studied the concept of CNN in rice leaf disease prediction and found that there is still scope for improvement in the overall performance of existing approaches. To address the limitations of previous works, we proposed a combination of the most recent squeeze-and-excitation (SE) blocks with a CNN-based Stacked parallel network (Das et al., 2021). According to our knowledge and belief, this approach has never been used before in classifying rice leaf diseases, automatically extracting the feature information and eliminating the invalid feature information. The following is a summary of this work's main contributions.

- We propose a novel architecture, a stacked parallel Convolution Neural Network with an added squeeze and excitation (SE) component SE_SPnet to effectively predict the disease of rice leaf in a natural environment.
- The proposed architecture is fine-tuned with multiple optimizers such as Adagrad, Adam, and Stochastic Gradient Descent with momentum optimizers with learning rates of 0.01 and 0.001 for the identification and detection of diseased rice leaf images.
- We compare the proposed model with three different standards, CNN VGG16, DenseNet121, and InceptionV3, for the Rice Leaf Disease Image datasets.

The remainder of the paper is organized as follows. Section 2 summarizes the related work. Section 3 provides the details of the materials used and methods proposed to address the given problem. Section 4 discusses the experimental setup details and presents results obtained through these experiments. Section 5 is a detailed discussion of the obtained results. Finally, Section 6 concludes our work.

2 | RELATED WORK

Recently, a number of researchers have introduced numerous CNN-based deep learning (DL) approaches in an effort to classify images of plant leaf disease with high accuracy. The primary advantage of deep learning over traditional machine learning methods is that it supports end-to-end learning and does not require the extraction of complicated handcrafted features. These methods have made significant contributions to image classification. The following three steps are often employed in the detection and classification of rice leaf diseases. The first step is segmentation, which divides the images into categories with related features for simplified analysis. The second step is the feature extraction process, which characterizes symptoms using features such as the shape and colour of the damaged part of the rice leaf, and the extracted feature is used to identify the diseases. Finally, the classification process to classify most of the diseases that appeared in the rice leaf (Lu et al., 2021). Table 1 summarizes related works with information on their objectives, methods, accuracy, and other observations and are discussed below:

TABLE 1 Summary of the related work

Reference, year	Objective	Proposed methodology	Result (accuracy)	Remarks
Hang et al., 2019	Reducing the long training convergence time and too-large model parameters of traditional convolution neural network	Combined inception module, squeeze-and-excitation (SE) module, and global pooling layer to identify diseases.	91.7%	• Despite optimization of the parameters, accuracy is not high.
Tang et al., 2020	Mobile devices compatible small and low-latency models	Improved ShuffleNet architecture applying the channel-wise attention (CA) mechanism	99.14%	• Slow convergence time and costlier. • High rate of accuracy and compressed model.
Jiang et al., 2020	Image recognition of four rice leaf diseases model that effectively extracts the implicit feature information to achieve a better recognition result	Mean shift image segmentation method, CNNs, and SVM	96.8%	• Model has many layers resulting large number of parameters to train. • Without the segmentation algorithm, accuracy might be lower. • Extra burden of manually design and extract features.
Bari et al., 2021	Real-time detection of rice leaf diseases architecture using real field rice leaf images	Faster R-CNN framework applies advanced regional proposal network (RPN) architecture to generate candidate regions very precisely	• Rice blast: 98.09% • Brown spot:98.85% • Hispa: 99.17% • Healthy rice leaf: 99.25%	• Accuracy is encouraging for infected leaves in both laboratory-based images and real-field images. • Not a fully automated system. • Time-consuming, to extract all objects algorithm needed numerous passes.
Chowdhury et al., 2021	Early and automatic disease detection of healthy and unhealthy tomato leaf images to reduce the adverse effects of diseases	• U-net and modified U-net • EfficientNet-B0 • EfficientNet-B4 • EfficientNet-B7	• Modified U-net: 98.66%, • EfficientNet-B7 six-class classification and binary classification: 99.12% and 99.95%, respectively • EfficientNet-B4 ten-class classification: 99.89%	• Network improved when trained with more parameters.
Chen et al., 2021	Identify rice plant diseases from real-life images	MobileNet-V2 paired with the attention mechanism	99.67%	• Datasets used in this study are small. • Generalization of the model is often poor.
Sharma, Nath, et al., 2022	Model for identification of nutritional deficiency in rice plant that can be used in an embedded system	Cloud-based ensemble averaging models employ transfer learning architectures	• InceptionResNetV2 (Kaggle dataset): 90%, • Xception (Mendeley datasets): 95.83%.	• Higher predictive accuracy but expensive in terms of both time and space
Our proposed model	A compatible model with moderate training convergence time, low cost and higher accuracy	Four parallel convolution layers stacked with an added squeeze and excitation (SE) element for effective prediction	99.2%	• Reduced the model parameters as well as training convergence time • High accuracy • Fully Automatic and reliable • Low cost

Hang et al. (2019) developed an improved CNN structure and used a sizable dataset of several plant leaf diseases to identify and classify leaf diseases. It is a conventional VGG16-based 5-layer convolutional model along with SE and Inception modules. The global average pooling layer substitutes the fully connected layer to reduce training time, parameter, and memory requirements and improve classification accuracy.

Tang et al. (2020) proposed a novel model to classify diseased grape leaves. This model incorporates the lightweight CNN ShuffleNet V1 and ShuffleNet V2 as the base model. The SE blocks have been used as a channel-wise attention mechanism, improving the base model's structure. The authors demonstrated that the proposed model performed better in classifying and detecting the disease, with an accuracy of 99.14% on the open PlantVillage dataset, which includes 4062 grape leaf images.

Jiang et al. (2020) applied the CNN model to extract the features from the images of diseased rice leaves. The authors used a support vector machine (SVM) to classify and predict the specific disease with different parameters. During the image segmentation stage, the mean shift image segmentation approach is used to separate lesions from images of rice leaf disease. Based on deep learning and SVM, this model gives an accuracy of 96.8%.

Bari et al. (2021) deploy a faster region-based CNN (Faster R-CNN) for rice leaf disease detection in real-time. The proposed model used advanced region proposal network (RPN) architecture that precisely identifies the location of the object to create candidate regions. With an accuracy of 99.17%, 98.09%, and 98.85%, the proposed model performed exceptionally well in detecting three different rice leaf diseases, hispa, rice blast, and brown spot, respectively.

Chowdhury et al. (2021) proposed a deep CNN-based on advanced EfficientNet, namely EfficientNet-B0, EfficientNet-B4, and EfficientNet-B7. global average pooling (GAP), the last layer of the CNN, is followed by a Dense layer with a size of 1024 and a 25% dropout. Lastly, the Softmax layer is used for the probability prediction of tomato leaf diseases. U-net and Modified U-net, two segmentation models, were used in the segmentation of leaves. The modified U-net segmentation model achieved an accuracy of 98.66%. EfficientNet-B7 outperformed other methods on the plant village dataset for both binary classifications using segmented images and six-class classifications, with an accuracy of 99.12% and 99.95%, respectively.

Chen et al. (2021) designed a novel network to identify several diseases in rice plants. The proposed network used the base network, MobileNet-V2, pre-trained on ImageNet and considered the attention mechanism Mobile-Atten. Furthermore, the model was trained using twice transfer learning, and the loss function was enhanced. On the public dataset, the proposed approach obtained an average identification accuracy of 99.67%.

Sharma, Nath, et al. (2022) studied the issue of the resource-constrained embedded system used in smart agriculture. To recognize the deficiencies in rice plants, they use six different transfer learning architectures, such as DenseNet201, ResNet50V2, Xception, InceptionResNetV2, InceptionV3, and VGG19. The proposed framework's optimum performance attained an accuracy of 95.83% with Xception and 90% with InceptionResNetV2 in the Mendeley and Kaggle datasets, respectively. Furthermore, model averaging-based ensembling of TL models was considered to generate 11 ensemble averaging models using VGG19, Xception, Inception-ResNetV2, and DenseNet201.

3 | MATERIALS AND METHODS

This section discusses the materials and methods we have used to implement the proposed system.

3.1 | Dataset

In this article, rice leaf disease image datasets from Mendeley are used (Sethy, 2020). We split the data set in the ratio of 80:20, with 80% for training and 20% for testing purposes. Moreover, 20% of the training data are kept as a validation set to avoid overfitting. There are a total of 4624 images in this dataset for rice leaves, divided into 1584 images for bacterial blight, 1440 images for leaf blasts, and 1600 images for brown spots. The image dimensions are reduced to 128 * 128 pixels for computational modesty. The dataset's sample images are depicted in Figure 1.

3.2 | Proposed model

To solve the majority of image recognition problems, the CNN model uses hyperparameters like the number of neurons in the fully connected layers, number of epochs, batch size, learning optimizer, learning rate, size of filters, number of filters in each layer, the total number of layers in the model, etc. In this work, we propose a SE_SPnet, a combination of the most recent Squeeze-and-Excitation blocks with a CNN-based Stacked parallel network, where Squeeze-and-Excitation blocks reshape the channel-specific features responses adaptively (Hu et al., 2018). Our proposed architecture of SE_SPnet is depicted in Figure 2.

(a) Bacterial blight (b) Leaf Blast (c) Brown Spot

FIGURE 1 Sample images of rice leaf diseases.

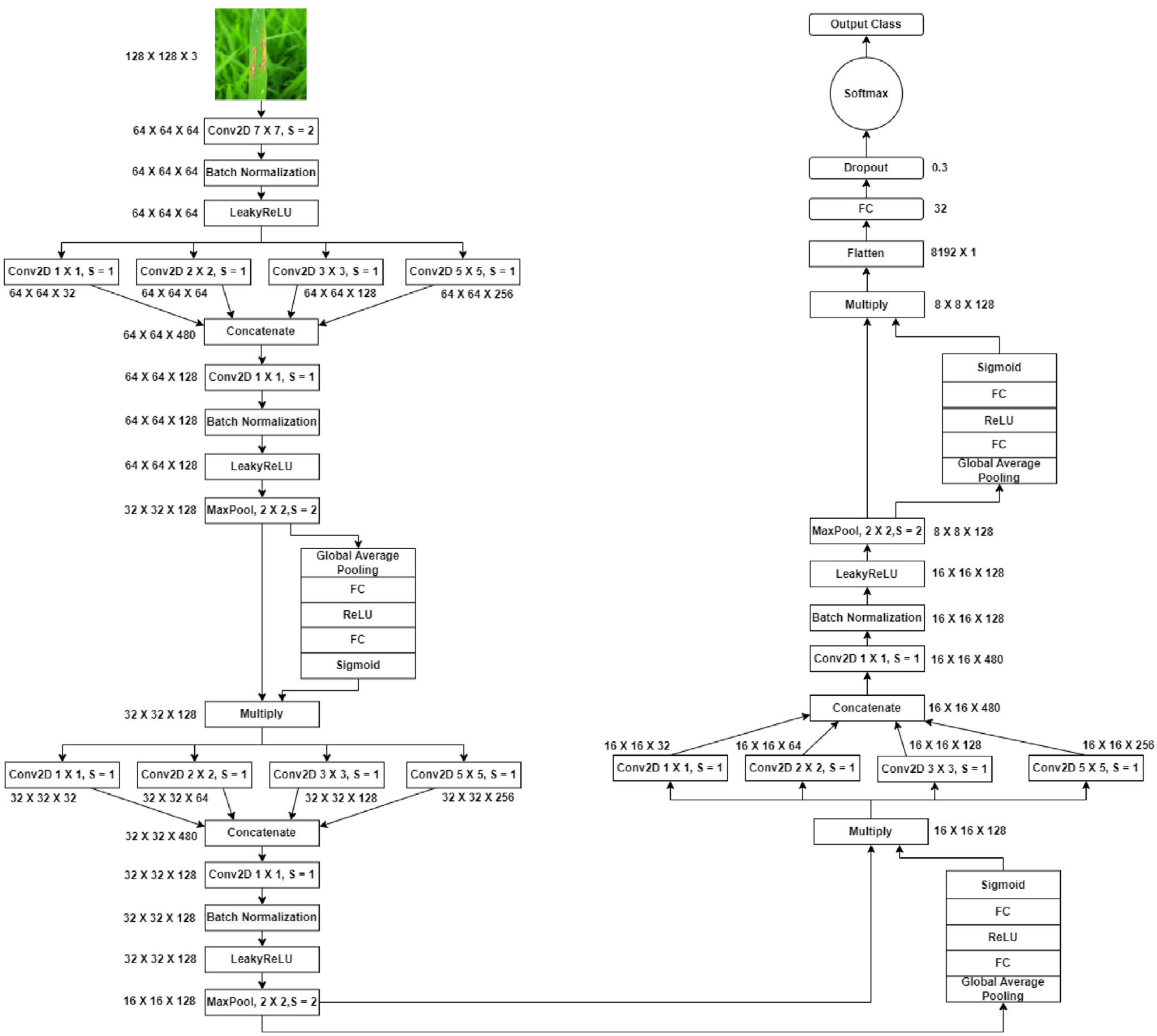

FIGURE 2 Proposed model (SE_SPnet).

First, the input layer, the first layer of the model, accepts the reshaped 128 × 128 three colour channels, R, G, and B images. The kernel size of 7 × 7 convolution layer is the first layer of SE_SPnet with 2 × 2 strides. The input image is resized using the first convolution layer. The following layers, batch normalization (BN) and LeakyReLU activation serve as transition layers. The transition layer regulates the input to expedite the training. The stacked parallel network block is comprised of four parallel convolution layers. Each layer used 1 × 1 stride with ReLU activation. For appropriate global and local feature abstractions, the kernel sizes of 1 × 1, 2 × 2, 3 × 3, and 5 × 5 convolution layers are stacked in a parallel way. The numbers of the kernels are arranged in ascending order of 32, 64, 128, and 256, respectively.

The concatenate block combines the output of four parallel convolution layers. It penetrates through the 1 × 1 convolution layer, followed by Batch Normalization, LeakyReLU activation, and 2 × 2 MaxPooling layer with strides 2 × 2. The output of the 2 × 2 MaxPooling layer is fed to the SE block. The Squeeze Excitation operation revamps the quality of convolution features produced by the network by accurately customizing the inter-dependencies between channels. The network is squeezed to prevent channel dependency. This is implemented by applying global average pooling across all input channels prior to reshaping, which makes use of contextual information that is present outside of the perceptron. The output from the squeeze operation is input to the excitation function. The excitation function captures the channel-wise dependencies. This is accomplished by sandwiching the reduction layer between two (FC) fully connected layers, then it is followed by a sigmoid activation layer. Before departing the SE block, the output from the original maxpooled is multiplied by the Squeeze and Excitation block output. The stacked parallel network block takes the multiplied output as input and so forth. The last convolution layer that is flatten. Lastly, the fully connected (FC), has 32 neurons. The output from the flatten is insert to the FC layer. Subsequently, a 30% dropout layer is added to prevent overfitting. For Multi class classification, softmax activation function used as the final layer. A more detailed and layer-wise description of the model's parameters, such as filter size and strides, activation maps, and the number of trainable parameters, is shown in Table 2.

The equation of the softmax function is given below.

$$P(y=i|X;\theta)=\frac{\exp\left(\theta_i^T X\right)}{\sum_{k=1}^{k}\exp\left(\theta_i^T X\right)}, \tag{1}$$

where x is input vector to the model, θ^i is feature vector's dimension x, y_i is the output values, The output probability of each class for each input x is given by $P(y=i|X;\theta)$.

4 | EXPERIMENTAL WORK

This section provides the experimental setup details and obtained results.

4.1 | Setup

We used Python 3.7.12 to implement and analyse the proposed model with existing well-known state-of-the-art networks for rice leaf disease prediction. The experiments were carried out with the Tensorflow (Abadi et al., 2016) Python package in a Kaggle GPU environment which is freely available. The dataset is initially split into 80–20 random splits, with 80% of the data being used for training and 20% being used for testing. Additionally, 20% of the training data is used as a validation set to counter the overfitting issue. A variety of optimizers such as Adagrad, SGD (with momentum), Adam, and learning rate settings of 0.01 and 0.001 are used to train the proposed model, as shown in Table 3. Accordingly, a total of six sets of experiments were carried out. The efficacy of the proposed model was analysed using the results obtained from 10 separate arbitrary splits on 100 epochs and patience of 50 epochs as early stopping. The classifier of the proposed SE_SPnet was tested and validated 60 times with six sets of experiments, and each set included 10 runs.

4.2 | Results

The confusion matrix has been used to examine how well the proposed approach responded. The confusion matrix used to test the rice leaf test dataset is shown in Figure 3. The confusion matrix was used to calculate the true positive, true negative, false positive, and false negative values. The efficacy of the proposed model is evaluated using accuracy, F1-score, precision, sensitivity, and specificity. Table 4 shows the equations used to calculate the performance metrics from confusion metrics.

The sensitivity and specificity metrics indicate how well a model predicts classes. The true positive rate or sensitivity defines the percentage of correctly predicted diseased leaf images by the classifier, and the true negative rate or specificity defines the percentage of accurately predicted disease-free leaf images by the classifier. These metrics are crucial in an automated disease prediction system. The model performance

TABLE 2 Summary of the layer-wise parameters for the proposed SE_SPnet model

Name of the layers	Filter size (FS) and strides (S)	Activations	Number of trained parameters
Input layer	–	$128 \times 128 \times 3$	0
Conv2D_0, BatchNorma_0, LeakyReLU_0	$FS = 7 \times 7, S = 2$	$64 \times 64 \times 64$	$9472 + 256 + 0$
Conv2D_1	$FS = 1 \times 1, S = 1$	$64 \times 64 \times 32$	2080
Conv2D_2	$FS = 2 \times 2, S = 1$	$64 \times 64 \times 64$	16,448
Conv2D_3	$FS = 3 \times 3, S = 1$	$64 \times 64 \times 128$	73,856
Conv2D_4	$FS = 5 \times 5, S = 1$	$64 \times 64 \times 256$	409,856
Concatenate_0 (Conv2D_1, 2, 3, 4)	-	$64 \times 64 \times 480$	0
Conv2D_5, BatchNorma_1, LeakyReLU_1	$FS = 1 \times 1, S = 1$	$64 \times 64 \times 128$	$61{,}568 + 512 + 0$
MaxPooling2D_0	$FS = 2 \times 2, S = 2$	$32 \times 32 \times 128$	0
GlobalAvgPool2D_0	–	$1 \times 1 \times 128$	0
Dense-FC_0	–	$1 \times 1 \times 8$	1024
Dense-ReLU_0	–	$1 \times 1 \times 8$	1024
Dense-FC_1	–	$1 \times 1 \times 128$	1024
Dense-Sigmoid_0	–	$1 \times 1 \times 128$	1024
Multiply_0	–	$32 \times 32 \times 128$	0
Conv2D_6	$FS = 1 \times 1, S = 1$	$32 \times 32 \times 32$	4128
Conv2D_7	$FS = 2 \times 2, S = 1$	$32 \times 32 \times 64$	32,832
Conv2D_8	$FS = 3 \times 3, S = 1$	$32 \times 32 \times 128$	147,584
Conv2D_9	$FS = 5 \times 5, S = 1$	$32 \times 32 \times 256$	819,456
Concatenate_1 (Conv2D_6, 7, 8, 9)	–	$32 \times 32 \times 480$	0
Conv2D_10, BatchNorma_2, LeakyReLU_2	$FS = 1 \times 1, S = 1$	$32 \times 32 \times 128$	$61{,}568 + 512 + 0$
MaxPooling2D_1	$FS = 2 \times 2, S = 2$	$16 \times 16 \times 128$	0
GlobalAvgPool2D_1	–	$1 \times 1 \times 128$	0
Dense-FC_2	–	$1 \times 1 \times 8$	1024
Dense-ReLU_1	–	$1 \times 1 \times 8$	1024
Dense-FC_3	–	$1 \times 1 \times 128$	1024
Dense-Sigmoid_1	–	$1 \times 1 \times 128$	1024
Multiply_1	–	$16 \times 16 \times 128$	0
Conv2D_11	$FS = 1 \times 1, S = 1$	$16 \times 16 \times 32$	4128
Conv2D_12	$FS = 2 \times 2, S = 1$	$16 \times 16 \times 64$	32,832
Conv2D_13	$FS = 3 \times 3, S = 1$	$16 \times 16 \times 128$	147,584
Conv2D_14	$FS = 5 \times 5, S = 1$	$16 \times 16 \times 256$	819,456
Concatenate (Conv2D_11, 12, 13, 14)	–	$16 \times 16 \times 480$	0
Conv2D_15, BatchNorma_3, LeakyReLU_3	$FS = 1 \times 1, S = 1$	$16 \times 16 \times 128$	$61{,}568 + 512 + 0$
MaxPooling2D_2	$FS = 2 \times 2, S = 2$	$8 \times 8 \times 128$	0
GlobalAvgPool2D_2	–	$1 \times 1 \times 128$	0
Dense-FC_4	–	$1 \times 1 \times 8$	1024
Dense-ReLU_2	–	$1 \times 1 \times 8$	1024
Dense-FC_5	–	$1 \times 1 \times 128$	1024
Dense-Sigmoid_2	–	$1 \times 1 \times 128$	1024
Multiply_2	–	$8 \times 8 \times 128$	0
Flatten	–	8192×1	0
Dense-ReLU	–	32FC	262,176
Dropout (0.3)	–	Dropout (30%)	0
Dense-Softmax	–	3	99

TABLE 3 Optimisers and parameter values used in the experiment

Parameter	Learning rate
Adagrad	0.01 and 0.001
SGD	0.01 and 0.001
Adam	0.01 and 0.001

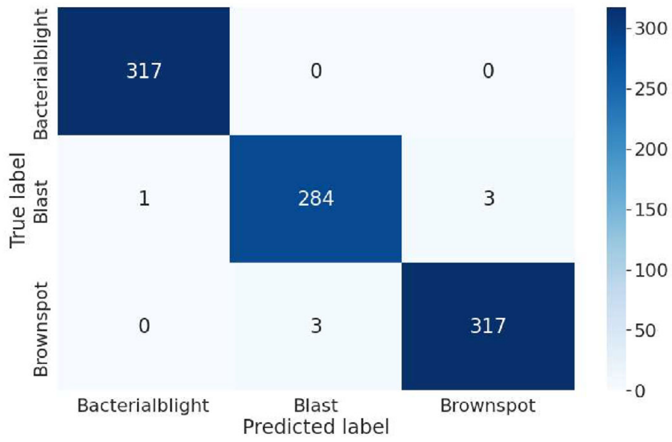

FIGURE 3 Confusion matrix of SE_SPnet.

TABLE 4 Performance metrics for rice leaf disease detection

Performance measure	Equation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Sensitivity or recall	$\frac{TP}{TP+FN}$
Specificity	$\frac{TN}{TN+FP}$
Precision	$\frac{TP}{TP+FP}$
F1-score	$2\frac{PrecisionxRecall}{Precision+Recall}$

in terms of sensitivity and specificity is represented by the bar graph in Figure 4. The sensitivity and specificity were evaluated with three different optimizers and two learning rate parameters for each optimizer. With a high sensitivity score, the proposed SE_SPnet performed magnificently. SE_SPnet achieves the highest sensitivity of 0.98. The specificity is comparatively lower than the sensitivity. Specificity and sensitivity have an inversely proportionate connection. As a result, the specificity decreases as the sensitivity increases, and vice versa. Our model with slightly lower specificity predicted positive without the disease due to the inter-class similarity nature of the dataset used.

The performance of the proposed model has been evaluated using error rate, precision, and recall metrics. The results against these performance metrics are listed in Table 5, which are obtained against optimizers Adam, Stochastic Gradient Descent (SGD), and Adagrad with the two learning rate parameters of 0.01 and 0.001. All sets of results are encouraging; in particular, SGD, with a momentum optimizer and learning rate of 0.01, achieved comparatively better results for every evaluation metric that has been considered.

The comparative analysis of proposed SE_SPnet with three different Standard CNN techniques, VGG-16, DenseNet121, and InceptionV3, is given in Table 6. The proposed SE_SPnet model obtained the best accuracy of 99.2% because our method gives the network a much better receptive field. The four convolutional layers stacked in parallel assist in analysing regions of interest more effectively, and the squeeze-and-excitation block cuts down on computational costs by removing unnecessary features. The proposed SE_SPnet has demonstrated outstanding results against other additional evaluation metrics of error rate, sensitivity, specificity, precision, recall, and F1-score.

The processing time is crucial in precision agriculture, enabling real-time operation. In addition to its performance comparison, the training time for the given dataset and the testing times for each image of the proposed model have also been compared against standard CNN models. Since our proposed model obtained better results in SGD with a momentum optimizer of a learning rate of 0.01, we have set the same learning

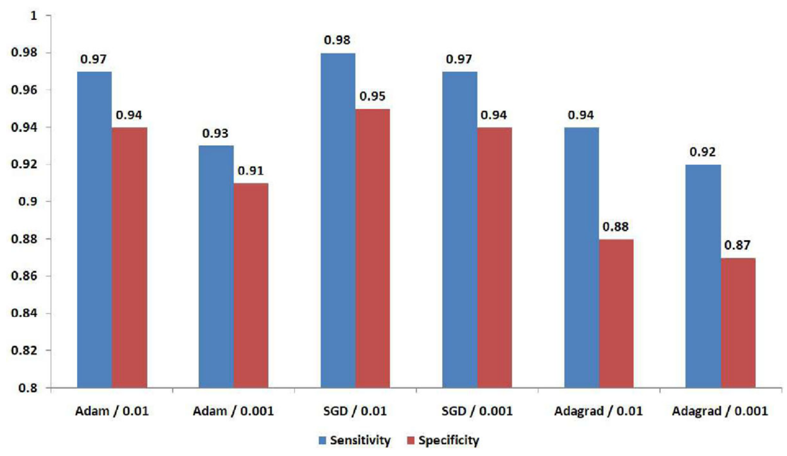

FIGURE 4 Bar graph of SE_SPnet in terms of sensitivity and specificity.

TABLE 5 Average precision, recall and error rate of the SE_SPnet

Optimizer	Learning rate	Precision	Recall	Error rate
Adam	0.01	0.957	0.971	0.046
	0.001	0.929	0.936	0.087
SGD	0.01	0.976	0.985	0.026
	0.001	0.962	0.978	0.047
Adagrad	0.01	0.913	0.941	0.096
	0.001	0.907	0.923	0.085

TABLE 6 Comparative analysis with standard CNNs

Network	Accuracy	Sensitivity	Specificity	Precision	Recall	F1-score
VGG16 (Simonyan & Zisserman, 2014)	0.934	0.923	0.949	0.936	0.923	0.945
DenseNet121 (Huang et al., 2016)	0.953	0.943	0.964	0.945	0.943	0.936
InceptionV3 (Szegedy et al., 2016)	0.917	0.913	0.925	0.919	0.913	0.903
SE_SPnet	0.992	0.982	0.985	0.984	0.982	0.985

TABLE 7 Comparison based on computational effort with standard CNNs

Network	Total training time (sec)	Testing time (millisecond/image)
VGG16 (Simonyan & Zisserman, 2014)	892	3.091
DenseNet121 (Huang et al., 2016)	839	3.184
InceptionV3 (Szegedy et al., 2016)	1148	4.259
SE_SPnet	779	1.846

rate for the VGG-16, DenseNet 121, and Inception-V3 models. The results were obtained from 10 separate arbitrary splits into 100 epochs and patience of 50 epochs as early stopping. Our proposed model is found to be much faster than the VGG-16, DenseNet 121, and Inception-V3 models on the same datasets, as shown in Table 7.

The performance of the proposed method is evaluated in comparison to other existing methods using the same dataset. The results are listed in Table 8. The comparison reveals that our proposed model outperforms most of these existing works. Our proposed model attained an accuracy of 99.2%, which is close to the deep learning-based CNN model proposed by Sharma et al. (2022).

TABLE 8 Comparison to other existing methods

Author and year	Method	Result
Ahmed et al., 2019	Decision tree algorithm	97% (testing accuracy)
Sethy et al., 2020	ResNet50 with SVM	98.38% (F1-score)
Jiang et al., 2020	Combination of deep learning and SVM model	96.8% (testing accuracy)
Sethy et al., 2021	VGG16 with SVM	97.31% (testing accuracy)
Sharma et al., 2022	Deep learning based CNN model	99.58% (testing accuracy)
Mohapatra et al., 2023	Custom-CNN	97.47% (testing accuracy)
Our work	SE_SPnet	99.2% (testing accuracy)

5 | DISCUSSION

Rice leaf diseases are a serious threat to the world's food supply. Image classification has become more convenient and sophisticated because of the rapid development of intelligent systems. Rice leaf diseases can be detected and identified automatically with great accuracy, which allows for accurate disease prevention and control. Specifically, AI-based methods are being investigated in-depth for diagnosing rice leaf disease. However, the accuracy of existing CNNs in predicting rice leaf diseases is low. Thus, by improving the SE_SPnet, the proposed CNN architecture based on stacked parallel Convolution Neural Network with squeeze-and-excitation block is proven to be an innovative solution to the task. The SE block extracts feature information automatically and removes any unnecessary features. With different optimizers, Adam, SGD (with momentum), and Adagrad, having learning rate settings of 0.01 and 0.001, the efficiency of the proposed SE_SPnet in disease classification is evaluated. The SGD (with momentum) optimizer on learning rate 0.01 produced better results among all the configurations taken into consideration against each evaluation metric. Experimental results show that SE_SPnet outperforms the other three standard CNN techniques and other existing image classification methods in detecting and identifying rice leaf diseases.

To evaluate the robustness of our proposed model, we used the Kaggle GPU environment to implement the VGG-16, DenseNet 121, and Inception-V3 models and measured the performance of standard CNN techniques with the same dataset. The following are the reasons that the proposed model outperforms standard CNN techniques. The convolutional layers stacked parallel one on top of the other allow the network to have a significantly more excellent receptive field, which facilitates handling regions of interest that are available in various sizes. The usage of several different kernel sizes improves disease detection at multiple scales. The SE_SPnet's predictive abilities are significantly improved by fine-tuning the optimizer settings for the specific rice leaf disease classification task. Furthermore, the squeeze excitation block is placed after each stage to reduce the computational cost by removing redundant features.

The minimization of model parameters as well as training convergence time, is the key benefit of our proposed model over the existing models. Stacking makes it easier to handle regions of interest that come in different sizes. The different kernel sizes improve disease detection on multiple scales. After each stage, the squeeze excitation block reduces the computational cost by removing unnecessary features.

Further, a few examples of incorrectly identified images from two classes are shown in Figure 5. The proposed SE_SPnet performed remarkably well in correctly predicting diseased classes, as indicated by the outstanding sensitivity and specificity results. On the other hand, the examination of the misclassified samples of all three classes illustrates that those samples are difficult to classify correctly. Interestingly, the majority of the inaccurately classified samples exhibit a concern regarding inter-class similarity. A model with a high sensitivity or recall score (0.982), as well as a high precision value (0.984), is referred to as an effective prediction system. Thus, the proposed method can be implemented to identify and predict rice leaf diseases accurately.

6 | CONCLUSION

In this article, we proposed the SE_SPnet, a CNN-based stacked parallel convolution layer that combines convolution, transition, and max pooling layers in an appropriate intermediate manner. Moreover, the squeeze-and-excitation block is placed between the max pooling layers and the stacked parallel network block to optimize a network's representational power by allowing it to conduct dynamic channel-wise feature recalibration. The architecture has been fine-tuned and trained for detecting and identifying diseased rice leaf images. The different filter sizes used in parallel convolution layers and intermediate layers between parallel blocks assisted in extracting the essential features from the input images. According to the experimental results, SE_SPnet achieves a state-of-the-art accuracy of 99.2% on the rice leaf disease image datasets even without preprocessing the input image or data augmentation. Our model SE_SPnet, trained using different optimizers and learning rate parameters, effectively outperforms some standard CNN architectures. Compared to other existing models, the SE_SPnet accurately predicted rice leaf diseases. Our proposed model can be applied to automate the early recognition of rice leaf diseases. As a result, this study could help in the automatic disease

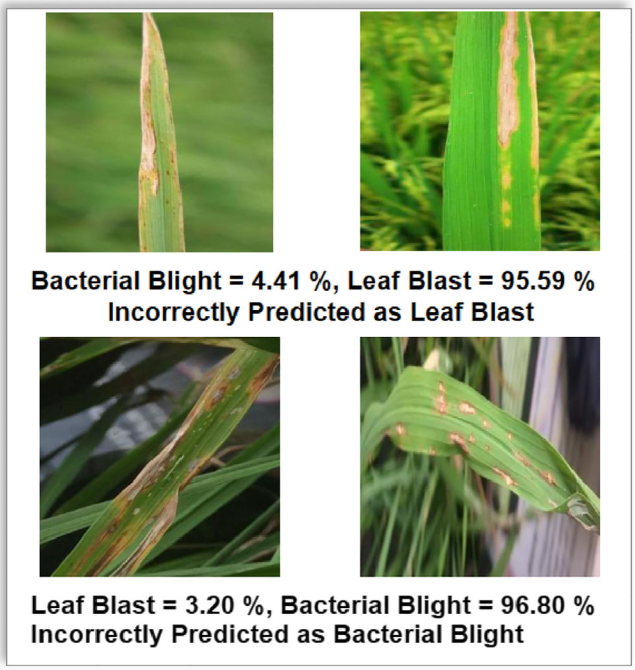

FIGURE 5 SE_SPnet incorrect prediction examples

identification of rice crops using cutting-edge technologies like cell phones, drone cameras, and robotic platforms at the early stage. Nevertheless, much larger and well-labelled data sets should be included in future research. In the future, we will concentrate on some other crop leaf disease types and the application of eXplainable AI (XAI) techniques for leaf disease classification. Also, we will focus on the improvement of the model so that it can recognize the disease of the images with real paddy field background at an early stage.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Rice Leaf Disease Image Samples at https://data.mendeley.com/datasets/fwcj7stb8r, reference number 10.17632/fwcj7stb8r.1.

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